SUB-MODULE 7.1 · FINE-TUNING FUNDAMENTALS

Fine-Tuning vs. Pre-Training

We will meet @ 11.05

Up to 8B → SLM

more than 8B → LLM

1000 → 9.5T → LLM.

Pre-training teaches language; fine-tuning specialises it — they operate at completely different scales

9 min	B3 – Apply	L2 – Intermediate	AI-FT-IN-TH-000003
Duration	Bloom's	Level	ID

Pre-Training vs Fine-Tuning — At a Glance

Pre-Training

- Train from random init on massive raw corpora
- Objective: next-token prediction (or masked LM)
- Dataset: trillions of tokens (Common Crawl, Books, Web)
- Cost: months of compute on 1000s of GPUs
- Result: general language understanding & generation
- Done once by the model provider (OpenAI, Meta, Anthropic)

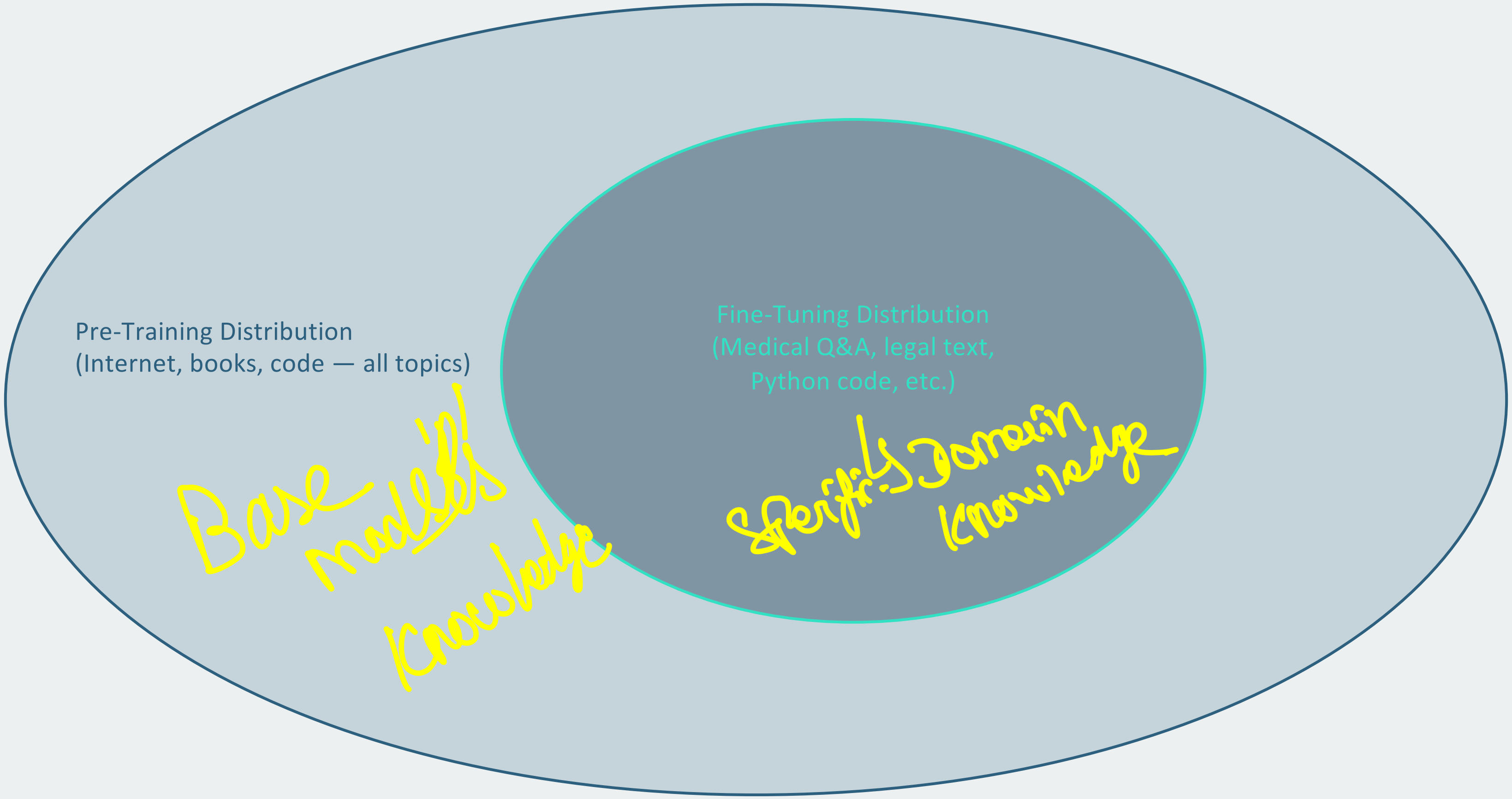
Fine-Tuning

- Continue training from a pre-trained checkpoint
- Objective: task-specific loss (instruction following, domain Q&A)
- Dataset: thousands to millions of curated examples
- Cost: hours to days on 1–8 GPUs
- Result: specialised behaviour on a target task
- Done by practitioners adapting the model for their use case

Scale Comparison — Pre-Training vs Fine-Tuning

Dimension	Pre-Training	Fine-Tuning
Training tokens	1T – 15T tokens	10K – 100M tokens
GPU hours	1M – 100M GPU-hours	10 – 10,000 GPU-hours
Cost estimate	$1M – $100M+	$10 – $10,000
Dataset type	Raw web, books, code	Curated instruction pairs
Who does it	AI labs (OpenAI, Meta...)	Practitioners, startups
Frequency	Once (rare updates)	Per task/domain/update cycle

Distribution Shift: How Fine-Tuning Narrows the Model



Fine-tuning shifts model probability mass from the broad pre-training distribution to the narrow target distribution. The closer the domains, the less data needed.

Two-Phase Fine-Tuning Workflow (Production Pattern)

1
Base Model
Llama 3, Mistral 7B, Qwen 2.5 — raw pre-trained weights

2
Phase 1: General Instruction
Fine-tune on FLAN / Alpaca-style datasets covering 60+ diverse tasks

3
Phase 2: Domain-Specific
Fine-tune on narrow domain data (medical, legal, code) — small, curated

4
Aligned Model
RLHF / DPO pass for safety and helpfulness (Sub-Module 7.4)

$0 \simeq n_1 < \; < \; < \; < n_{23} < n_{24}$

$X \to L1 \to L2 \to L3 \to \cdots \to L24 \to H \to \Sigma \to O$

Class – 1 to 6th – level

7th – 12th syntax

Semantics

world knowledge

What Pre-Training Gives You — The Representation Tiers

Layer 1–3: Surface Patterns

Morphology, punctuation, basic token statistics. Low-level n-gram patterns.

Layer 4–8: Syntax

POS tags, parse trees, co-reference. Grammar-level understanding.

Layer 9–16: Semantics

Word meaning, entity types, coreference resolution, factual associations.

Layer 17–24: World Knowledge

Factual knowledge, reasoning chains, domain facts. Highest-level representations.

Fine-Tuning Insight

Lower layers rarely need updating — they hold general skills. Upper layers specialise fastest.

Practical Rule

Fine-tuning on a closely related domain → needs fewer examples. Distant domain → needs more.

KEY REFERENCES

- Brown, T. et al. (2020). Language Models are Few-Shot Learners. NeurIPS 2020.
- Wei, J. et al. (2021). Finetuned Language Models Are Zero-Shot Learners. ICLR 2022.
- Touvron, H. et al. (2023). Llama 2. arXiv:2307.09288

▶ **Next: Concept 4: Catastrophic Forgetting (AI-FT-IN-TH-000004)**